

Element A

The Nashville Zoo is a long-standing organization that gives a home to many different animals. They have passed us the students of Brentwood High School to build them a multitude of things that allow for their Andean Bears to be fed, have proper mental nourishment, and allow for the enclosure workers to not have to worry about how and when the Bears get fed. Our group's task was specifically to automate a food launcher to allow for the Andean Bear enclosure workers to not have to step into the enclosure to feed the Bears. This will allow for the feeders to do other tasks during the day and will give the Bears proper nourishment as they are forage feeders. In this role as forage feeders, they need to be mentally asked with something almost daily so that they can get their energy out this is part of our duty in our part of our criteria to launch food in random spots and at random times during the day. It is vital that we are able to complete our task as if we fail to meet all our design criteria the zoo will have to go back to manually feeding the Bears which is a detriment to both the stakeholders of the Bears themselves and the zoo employees.



The two primary stakeholders of this project include the Andean Bears and their handlers. For both stake holders they have a different priority of which part of our automation project is more important. The bears care more about the randomized landing, so they can forage and work their brains. The bear handlers also care for the bears well being, but automating the shooting of the launcher is their priority number one. One secondary stakeholder would be persons who go to the zoo and watch the bears. They fall under secondary stakeholders because although they don't have a direct investment into the future of our project. If the bears are being mentally challenged they become more calm and are able to be better attractions for the zoo. Another secondary stakeholder would be other animal handlers in the facility. Because our job is to automate the launching mechanism it would allow for the bear handlers who are busy during the day, feeding the bears, to have more time to work with other enclosures.

The deadline for the final prototype is April 27th, where we will present the automated launcher prototype to the stakeholders. After our presentations they gave us feedback about what we can improve or need to take into consideration. Then we will have until May 16th to have the final product to show the zoo

